Chapter 59 Imaging using Mirrors
Calculus Physics II

Object – is the source of the light rays either by the object emitting light itself or by the light being reflected off the object.

Light rays are emanating in all directions from each point on the object, for example a tree that is illuminated by the sun has light reflecting off of it at each point on the tree that the light is incident and in all directions.

Self-Illuminating object – The object is seen because the object is actually emitting light such as a light bulb, exit sign or TV screen.

Objects that reflect light – The object is seen because light is being reflected off of the object, it is not a source of light but only reflects light.

Self-illuminating Exit sign



Object reflecting, tree and light source is the sun

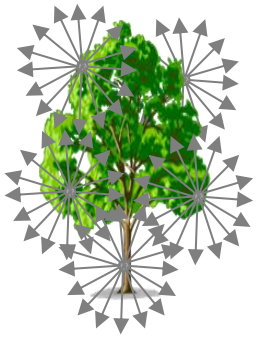
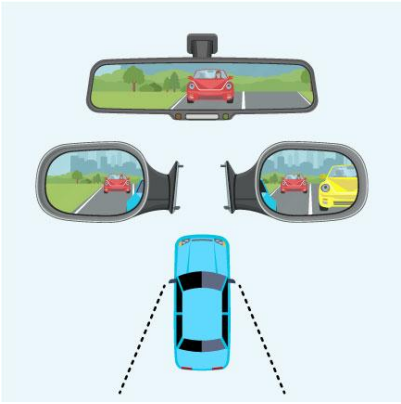


Image – The appearance of a reproduction of the object that your eye sees, it is formed when the light rays are reflected by a mirror or refracted by a lens. It is technically defined as the collection of focus points where light rays originating from an object either actually meet or appear to meet after passing through an optical system.

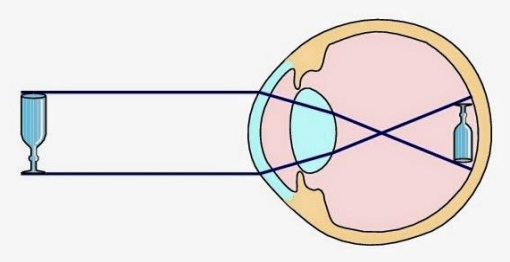
An image is the representation of the object due to the light from the object being reflected or refracted. It is not the light transmitted directly from the object to the eye but light that has been through some optical system. Images of objects are formed by lenses and mirrors.

The object is the red car in the mirror that is in back of the blue car, but we see the red cars image in the mirror.



- ✓ The image of an object is the mapping of the light from that object point by point keeping the same relative dimensions and relationships in the image as they are in the object.

- The eye/brain sees the object due to light rays entering the eye from the object and the image is formed by the eye.
- The eye/brain cannot tell the difference between the object or the image of the object.
- The eye itself (a lens) produces an image of what it sees on the retina at the back of the eye, the brain translates this into vision.



Mirrors – are smooth surfaces that reflect light in a predictable manner. The object and image for both mirrors and lens are characterized by the following

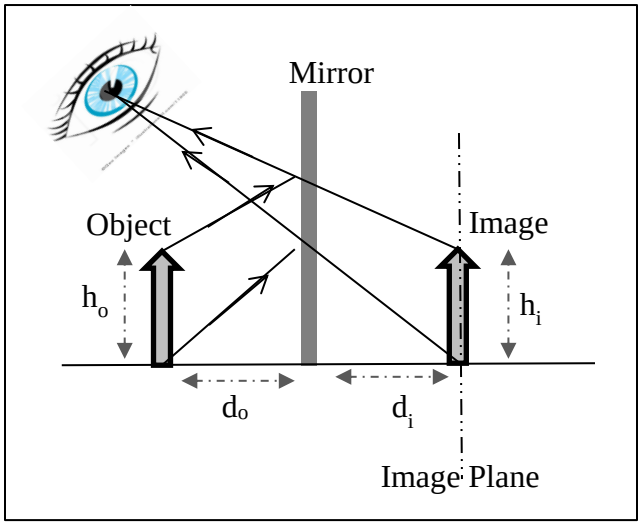
Object Distance (s_o or d_o) – The distance from the mirror to the object that is being imaged.

Image Distance (s_i or d_i) – the distance from the mirror to where the image appears in focus.

Object height (h_o) – the actual height of the object.

Image Height (h_i) – the height that the image appears to have.

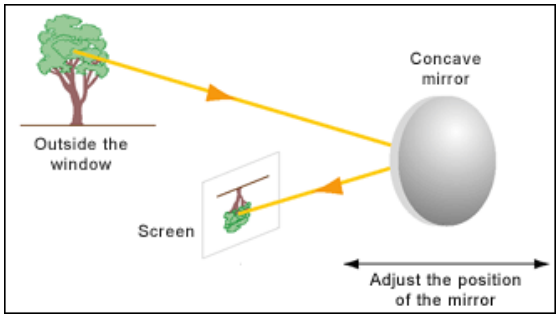
Image Plane – Imaginary plane in which the image appears in focus.



Linear Magnification (m) – the ratio of the image height to the object height and indicates the change in sizes of the image relative to the object.

Real Image – An image from which rays of light actually emanate (come from), the image can be seen on a screen if placed at the image plane.

Light from the tree (object) is focused on the piece of paper to form a real image of the tree. Light from each part of the tree travels to the mirror, reflects and forms the image of the tree on the paper. The image of the tree can be seen on the piece of paper due to the light rays reflecting off the paper.

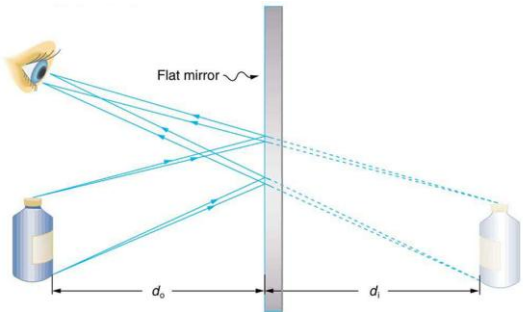


- ► It is the lens or mirror that creates the image of the object, if the lens were removed there would be no image formed on the paper.
- ► The cornea and lens of our eye forms a real image on the retina by using the cornea and lens.

Virtual Image – an image from which rays of light do not actually come, but only appear to do so, no image will appear on screen if placed at the image plane. A virtual image is not where it seems to be, it only appears to be at the location.

- ❖ Your eye cannot tell the difference between a real or virtual image.
- ❖ Must look into the optical system to see a virtual image.

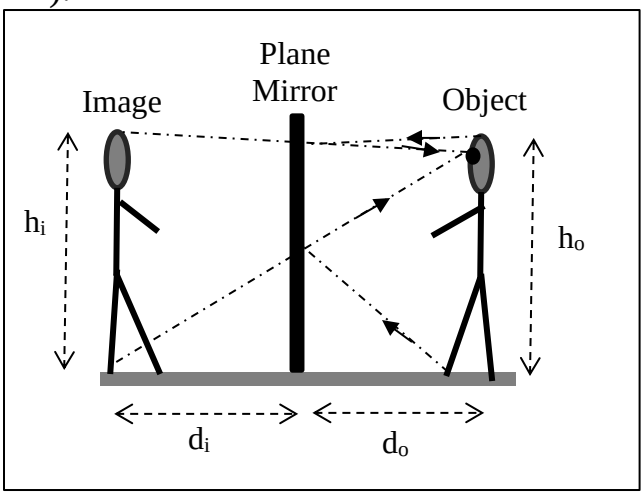
Light from the bottle reflects off the mirror to the eye of the person because our mind assumes that light travels in a straight line, the person will see the image of the bottle as if it is on the other side of the mirror, but the image is not really on the other side of the mirror it just appears to be there. If we place a piece of paper on the other side of the mirror, we will not see the image on the paper!



Plane Mirror – A flat reflecting surface which forms an upright, virtual image that is located as far behind the mirror as the object is in front of the mirror. A plane mirror has no curvature, so the image has the same proportions as the object.

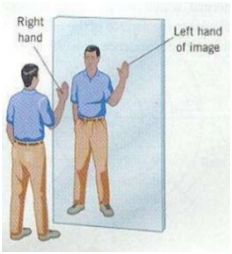
Image from Plane Mirror

1) Upright (erect) – same orientation as the object, the image is not upside down.
2) Image is the same size as the object ($m = h_i/h_o = 1$).
3) Image appears as far behind mirror as it is in front of mirror ($d_i = d_o$).
4) Image is virtual (light does not actually come from behind the mirror, it only appears to).
5) Image is reversed (your right hand is imaged left, finger pointing into mirror, but image points out)



- ✓ When you walk towards a mirror your image walks towards you at the same velocity, the relative velocity of you and your image is 2 x your velocity.
- ✓ In order to focus on the image of your face in the mirror you must focus on your face which is twice as far away as the mirror.
- ✓ The image height is the same as the object and does not change with object distance.

Lateral inversion in plane mirrors - Words and images appear backwards in the horizontal direction due to the fact that the image is reflected straight back. Your right arm appears in the image as your left or words are spelled with letters in reverse order.

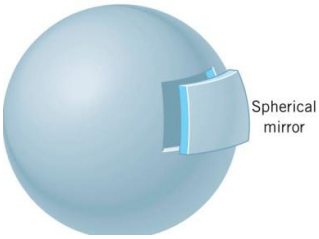


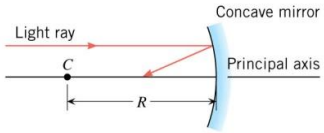


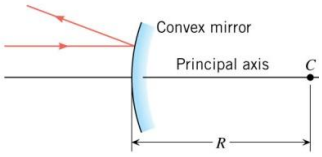
Curved Surface Mirrors – An object placed in front of a curved mirror will form an image of that object. The image formed depends on where the object is placed relative to the mirror. The curvature can be either convex (outward) or concave (inward).

Spherical mirror – A reflecting surface that has the shape of a section from the surface of a sphere, inside or outside surface.

Concave and Convex Mirrors – are formed by looking at sections of a polished sphere







Spherical Concave Mirror – The inside portion of the spherical surface is reflecting, and parallel light incident of a concave mirror converge the light to a point. The concave mirror will form both real and virtual images

Spherical Convex Mirror – The outside portion of the spherical surface is reflecting, and parallel light incident of a concave mirror diverges the light and appears to originate from a single point. The convex mirror only forms virtual images.

Center of Curvature (C) – is the point at the center of the sphere and the spherical surface. It is the radius of the spherical surface.

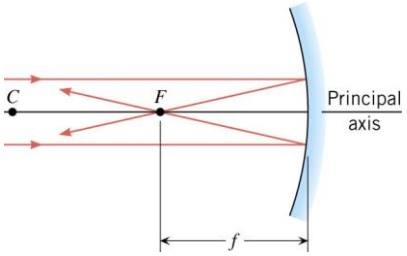
Radius of Curvature (R) – the distance from the center of curvature of the mirror to the surface of the mirror. Some books use C as the radius.

Principal axis – The straight line drawn through the center of curvature and the middle of the mirror's surface and is used as a reference for the object and image.

Paraxial rays – Rays that lie close to the principle axis.

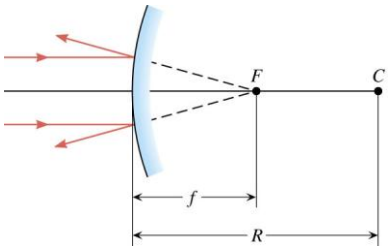
Focal point (f) for a convergent (concave) mirror – A point on the principal axis where paraxial rays that are parallel to the principal axis converge at a single point after being reflected from a concave surface.

- For a convergent mirror the light rays coming in parallel to the principal axis converges to the focal point on the same side of the mirror as the light.
- For a point source placed at the focal point the light rays will exit the mirror parallel.



Focal point (f) for a divergent (convex) mirror - is the point on the principle axis from which paraxial rays appear to emanate from rays that is incident parallel to the principal axis and diverge after being reflected from a convex surface.

- For a divergent mirror the light rays coming in parallel diverge so that if the reflected rays were traced back, they would converge at the focal point on the opposite side of the mirror as the light.



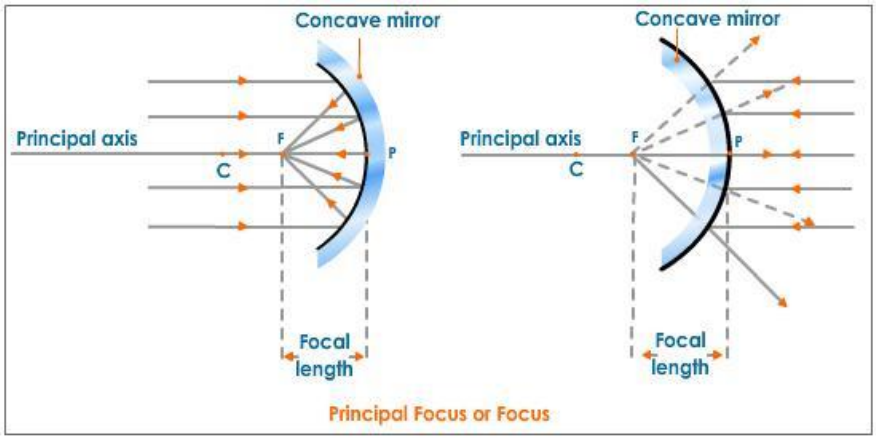
Focal Length (f) – The distance from the focal point to the surface of the mirror along the principle axis.

$f = \frac{1}{2} R$ for convergent/concave mirror

$f = -\frac{1}{2} R$ for divergent/convex mirror

- The "-" minus sign indicates the focal point is on the opposite side as the light rays and R is the radius of curvature of the mirror. This is a virtual focal point, it appears to be at a specific location but is not at that location.

- There is only one focal length for each mirror and this length defines how the mirror will image light incident on it.
- The focal length is defined by parallel light rays incident on the mirror.



Linear Magnification (m) – Measures the change in size of the image relative to the object. The ratio of the image height to the object height.

$$m = \frac{h_i}{h_o} \quad \text{therefore the height of the image is ginven by } h_i = mh_o$$

Using the object and image distances the magnification is also given as

$$m = -\frac{d_i}{d_o}$$

m is "+" for an image that is upright with respect to the object.
m is "-" for an image that is inverted (upside down) with respect to the object.

- Magnification depends on the focal length of the mirror.
- Magnification depends on where the objects is placed relative to the focal point.

m > 1 image larger than object
m = 1 image same size as object
m < 1 image is smaller than object

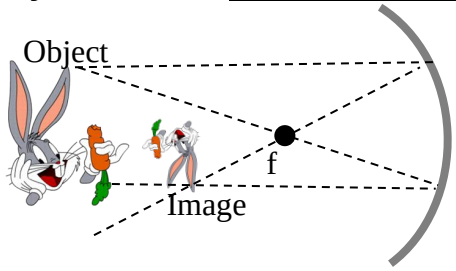
m is negative then image inverted
m is positive then image is upright

Example: When Bugs Bunny looks at himself in the mirror, assume he starts very far from the mirror.

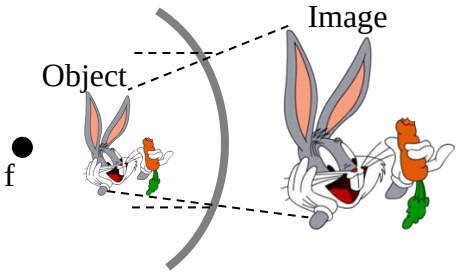
A) For a convergent mirror, when he (the object) is very far from the mirror the image is formed close to the focal point is real and inverted because the rays from the object are parallel. As he gets closer to the focal point his image gets larger, is still real and inverted and the image forms father away from the focal point. When he is at the focal point there is no image.

B) For a convergent mirror, when the object is placed inside the focal point the image that is formed is upright, larger and virtual. The image is larger the closer the object is to the focal point.

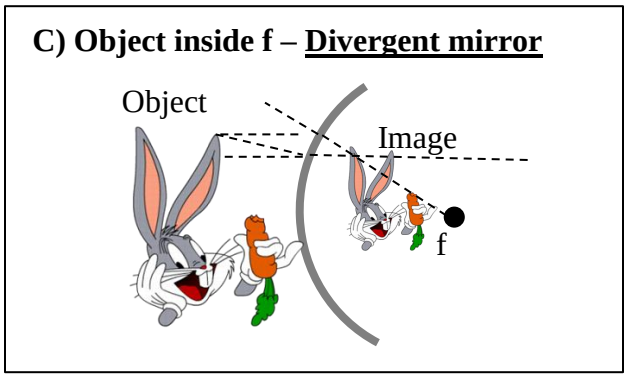
A) Object outside f– Convergent mirror



B) Object inside f – Convergent mirror

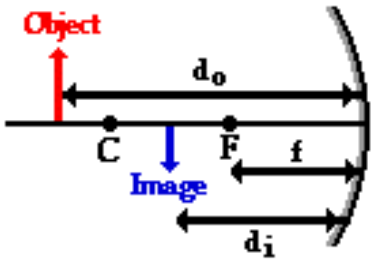


C) For a divergent mirror the object will always form an image that is smaller, upright and virtual.



Mirror (lens) equation – The mathematical relationship between the focal length (f) of the mirror, the distance the object is from the mirror (d_o) and the distance the image appears from the mirror (d_i).

$$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$$



d_o – object distance*
d_i - image distance*
f – focal length* of mirror

* All lengths are measured from the mirror's surface along principle axis.

Solving the equation for the image distance given focal length and object distance.

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]}$$

Sign Rules for Mirror's Object and Image distances

Object distance

\+ for object on the same side as of reflecting surface as incoming light ray. The object distance is always positive for a single mirror/lens.
\- for object on opposite side of reflecting surface as incoming ray, this can only happen in multiple imaging elements.

Image distance

\+ for image on the same side of the reflecting surface as the object (real image).
\- for image on opposite side of reflecting surface as the object (virtual image).

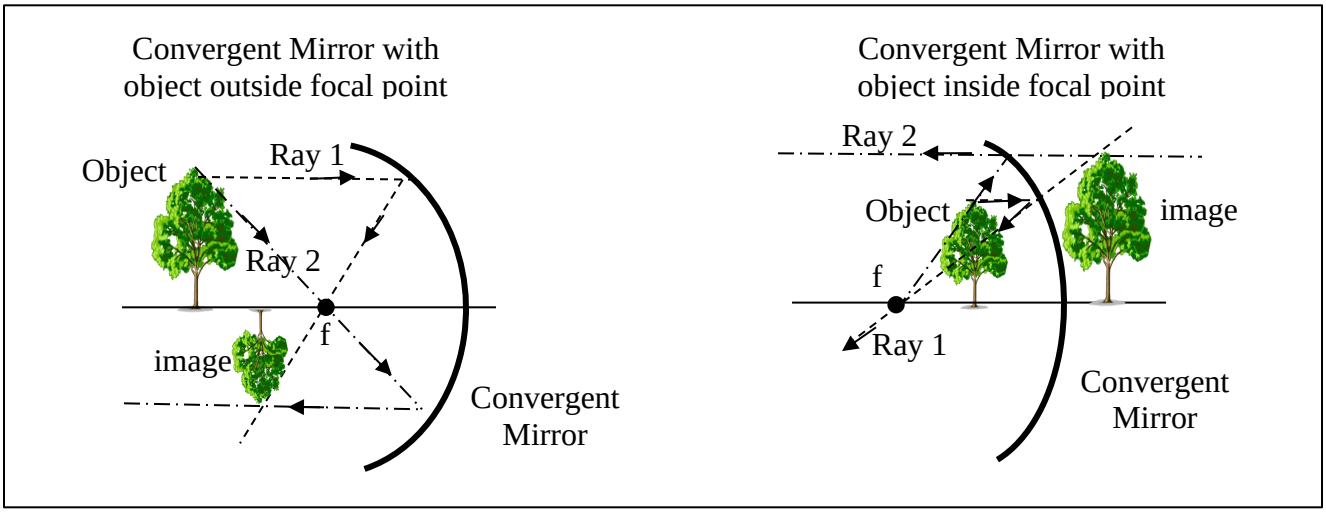
- For a Convergent mirror the real image is formed on the same side of the mirror as the object because the light rays are reflected, and the rays of light converge to form an image on the same side as the object.
- For lens the real image is formed on the opposite side of the lens as object because the light rays are transmitted through the lens and then converge to form the image on the opposite side of the lens than the object.

Ray Tracing – is a method to determine properties of the image by tracing two light rays from a single point on the object.

Ray tracing for a Convergent Mirror – using a point on the top of an object (tree below).

1) Ray # 1 - rays parallel to axis are reflected through the focal point (f).
2) Ray # 2 - rays through focal point are reflected parallel to the axis.
3) Where these rays intersect is the tip of the image of the tree.

(a) For a real image intersection of the reflected rays (1 and 2) gives distance and orientation of image from principal axis.
(b) For a virtual image the reflected rays do not converge so the reflected rays (1 and 2) are traced back behind the mirror to find their point of intersection.

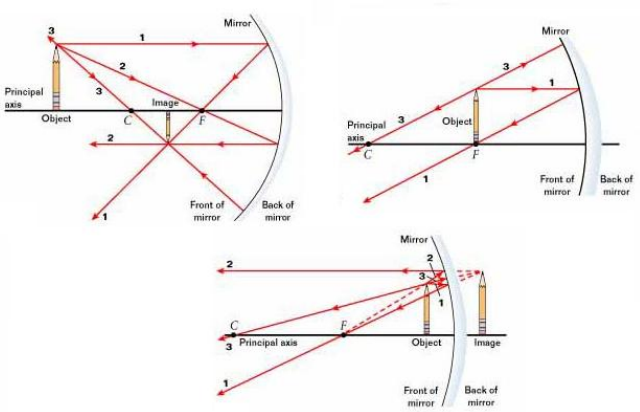


Object and Image for Convergent/Concave Spherical Mirror (Ray Tracing)

(a) For an object place very far away from the mirror (at infinity) the image will be formed at the focal point and will be inverted, real and smaller than the object.
(b) For an object placed outside of R, the image will be formed outside of the focal point and will be inverted and smaller than the object.
(c) For an object placed at R, the image will be formed at R, will be inverted and the same size as the object.

(d) For an object placed between R and f, the image will be formed outside of R and is real and bigger than the object.

(e) For an object placed at the focal point, there will be no image formed because the light coming off the image will not converge but stay parallel.



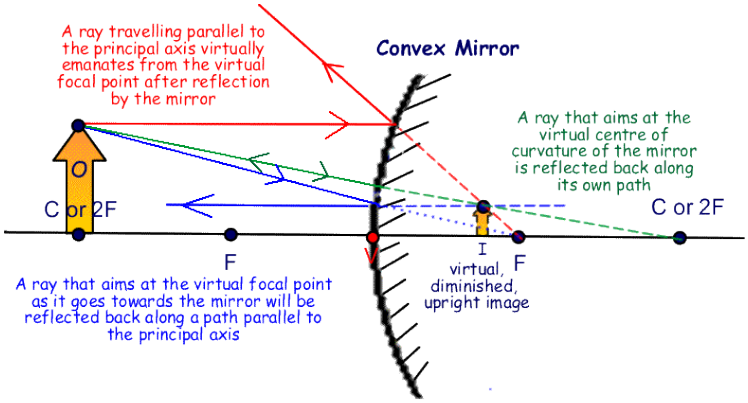
(f) For an object placed between the focal point and the mirror, the image will be virtual and appear to form behind the mirror, the image will be upright and larger than the object.

<u>Object and Image for Divergent/convex Mirror (Ray Tracing)</u> – by tracing the **<u>reflected</u>** rays back to where they intersect the location of the image can be found.

- ► For divergent mirrors the image is always virtual and smaller than the object.
- ► For a divergent mirror the magnification will always be less than 1.

For a point on the top of an object (the arrow)

1) Ray # 1 - rays parallel to axis are reflected as if they were coming from the focal point (f), trace the reflected ray back through the focal point on the other side of the mirror.
2) Ray # 2 – trace the ray as if it was going through focal point on the opposite side of the mirror and the reflected ray is parallel to the axis, trace the reflected ray back on the opposite side of the mirror as the object.
3) Ray # 3 - ray through radius of curvature (C) are reflected back on them selves
4) Where the rays intersect on the opposite side of the mirror as the object is the top of the image.





Summary of Sign convention for Spherical Mirrors

Focal Length

f is "+" for a convergent/concave mirror
f is "-" for a divergent/convex mirror

Object distance (d_o)

d_o is "+" if the object is in front of the mirror (real object)
d_o is "-" if the object is behind the mirror (virtual object)

Image distance (d_i)

d_i is "+" if the image is in front of the mirror (real image).
d_i is "-" if the image is behind the mirror (virtual image).

Magnification

m is "+" for an image that is upright with respect to the object
m is "-" for an image that is inverted with respect to the object

Examples: For an object (h_o = 5 cm) placed 100 cm in front of a convergent mirror with a focal length of 10 cm, what are the properties of the imaged formed?

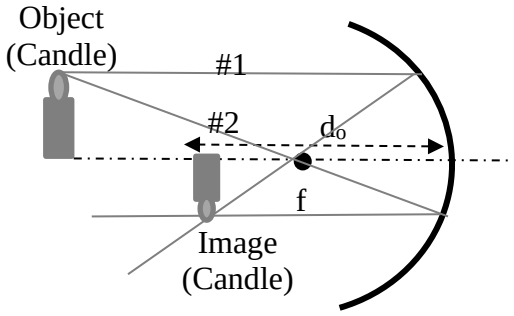
1) Calculate image distance:

$$\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$$

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\ cm} - \frac{1}{100\ cm}\right]}$$

d_i = + 11.1 cm

Convergent mirror – For object outside focal length



✓ A positive image distance means that the image is real

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{11.1\ cm}{100\ cm} = -0.11$$

- ✓ The negative on the magnification indicates the image is inverted relative to the object, the image is upside down.
- ✓ The magnification is less than 1 therefore the image is smaller than the object by a factor of 0.11.

3) Calculate the image height:

$$h_i = \ mh_o = (0.11)(5\ cm) = 0.55\ cm$$

- In summary the image is real, inverted, smaller (h_i = 0.55) than the object and formed 11.1 cm from the mirror and on the same side of the mirror as the object.

<u>Example</u>: For an object (h_o = 5 cm) placed 12 cm in front of a convergent mirror with a focal length of 10 cm, what are the properties of the imaged formed?

1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\ \text{cm}} - \frac{1}{12\ \text{cm}}\right]}$$

d_i = + 60 cm

- ✓ A positive image distance means that the object is real.

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{60\ cm}{12\ cm} = -5$$

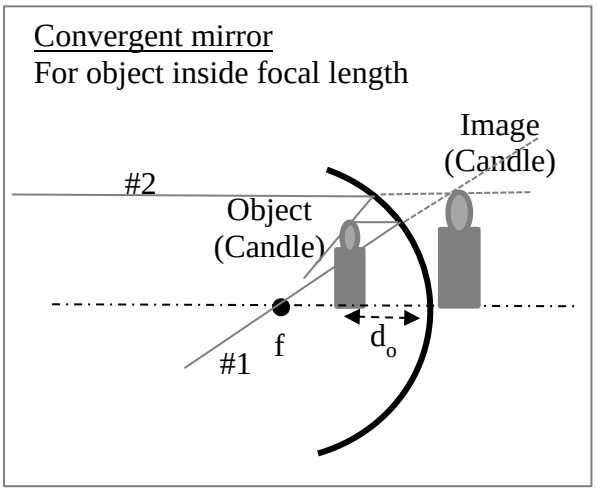
- ✓ The negative on the magnification indicates the image is inverted relative to the object, the image is upside down.
- ✓ the magnification is greater than 1 therefore the image is larger than the object by a factor of 5.

3) Calculate the image height:

$$h_i = \ mh_o = (5)(5\ cm) = 25\ cm$$

- In summary the image is real, inverted, larger (h_i = 25 cm) than the object and formed 60 cm from the mirror on the same side of the mirror as the object.

Example: For an object (h_o = 5 cm) placed 6 cm in front of a convergent mirror with a focal length of 10 cm, what are the properties of the imaged formed?



1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{10\ \text{cm}} - \frac{1}{6\ \text{cm}}\right]}$$

d_i = - 15 cm

- ✓ A negative image distance means that the image is virtual and appears to be on the opposite side of the mirror as the object.

2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{-15\ cm}{6\ cm} = +2.5$$

- ✓ The positive on the magnification indicates the image is upright relative to the object.
- ✓ the magnification is greater than 1 therefore the image is larger than the object by a factor of 2.5.

3) Calculate the image height:

$$h_i = mh_o = (2.5)(5\ cm) = 12.5\ cm$$

- In summary the image is virtual, upright, larger than the object (h_i = 12.5 cm) and appeared to be formed 15 cm on the opposite side of the mirror as the object.

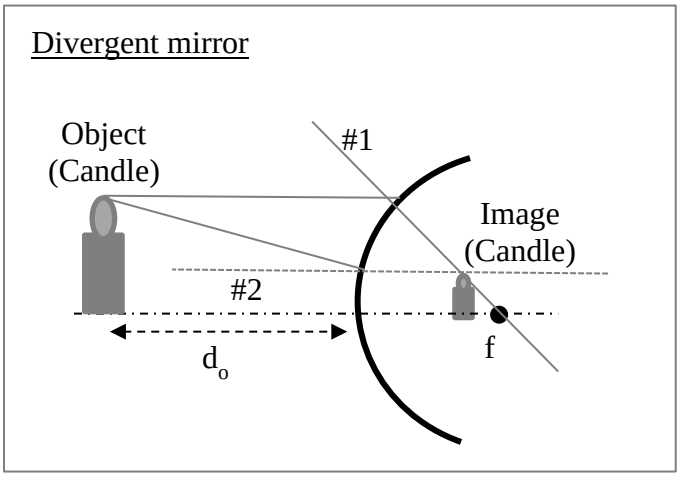
Example: For an object (h_o = 5 cm) placed 5 cm in front of a Divergent mirror with a focal length of negative 20 cm, what are the properties of the imaged formed?

1) Calculate image distance:

$$d_i = \frac{1}{\left[\frac{1}{f} - \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{-20\ \text{cm}} - \frac{1}{5\ \text{cm}}\right]}$$

d_i = - 4 cm

- ✓ A negative image distance means that the image is virtual.



2) Calculate magnification:

$$m = -\frac{d_i}{d_o} = -\frac{-4\ cm}{5\ cm} = +0.8$$

- ✓ The positive on the magnification indicates the image is upright relative to the object.
- ✓ the magnification is less than 1 therefore the image is smaller than the object by a factor of 0.8.

3) Calculate the image height:

$$h_i = mh_o = (0.8)(5\ cm) = 4\ cm$$

- In summary the image is virtual, upright, smaller than the object (h_i = 4 cm) and appeared to be formed 15 cm on the opposite side of the mirror as the object.

Example: Cathy notices that her image is formed 32 cm at what appears to be in back of the mirror and her image is 4 times larger and upright, how far is Cathy from the mirror (do = ?), what is the focal length of the mirror and what type of mirror?

d_i = - 32 cm - in back of the mirror means it is virtual therefore is negative
m = + 4 - it is 4 times larger and upright means the magnification is positive

$$m = -\frac{d_i}{d_o} \quad therefore\ d_o = -\frac{d_i}{m} = -\frac{-32cm}{4} = +8\ cm$$

$$f = \frac{1}{\left[\frac{1}{d_i} + \frac{1}{d_o}\right]} = \frac{1}{\left[\frac{1}{-32\ \text{cm}} + \frac{1}{+8\ \text{cm}}\right]} = +10.67\ cm$$

Plus sign means it is a convergent mirror.